

Designing an AI architecture for large scale, qualitative research and product strategy.

Source: /work/hbs-ai-institute/

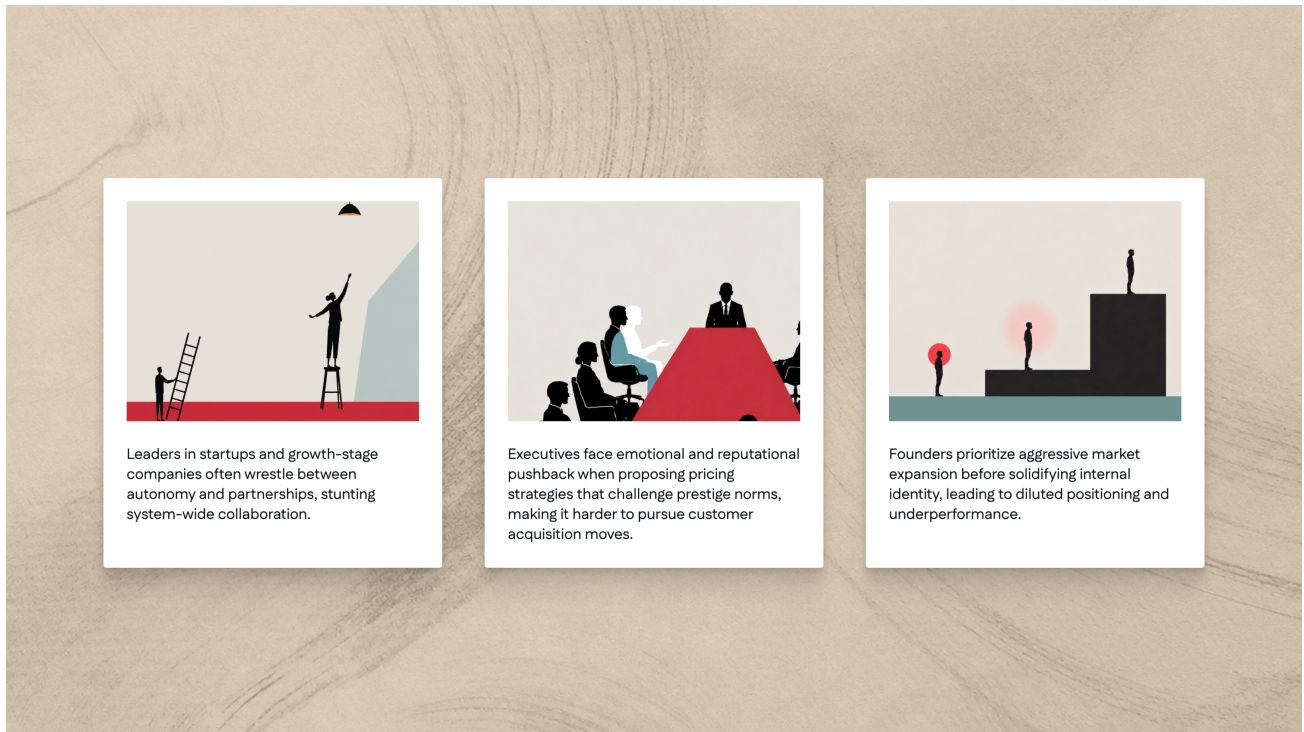


Image | Alt text: Designing an AI architecture for large scale, qualitative research and product strategy. | Source asset: /assets/hbs/hbs-cover.webp

As AI products multiplied around automation and content generation, HBS wanted to explore a more meaningful opportunity for AI in strategy among executives. The goal was to identify potential users and problems worth pursuing. The work resulted in:

- A defined problem set, frameworks to interpret and prioritize them, and follow-on product discovery.
- Human–AI interaction principles used to fine-tune model responses across HBS projects for business executives.
- A research architecture to quantify qualitative responses at scale for future research projects.

Stakeholder: HBS AI Institute (Previously D³ Institute)

Skills: Research StrategyProduct DiscoveryQualitative Data Analysis

Year: 2025 (4 weeks)

Role: Designer + Researcher • Team

The team

The HBS AI Institute engaged us to lead the research and design work, with the four team members below serving as project advisors.

- Tanya Flint Head of AI
- Yogesh Kumar Sr. Data Engineer | Advisor for project
- Jean-Luc Jackson Sr. Data Scientist | Advisor for project
- Rem Koning Assistant Professor | Advisor for project

Problem

There was no defined industry, executive role, or problem space. “Executives” could mean anyone from a CEO/Chairman to Director/Lead. “Strategy” could include almost every consequential decision they made.

Without clear boundaries, almost any polished answer could appear complete, even when important situations had been missed.

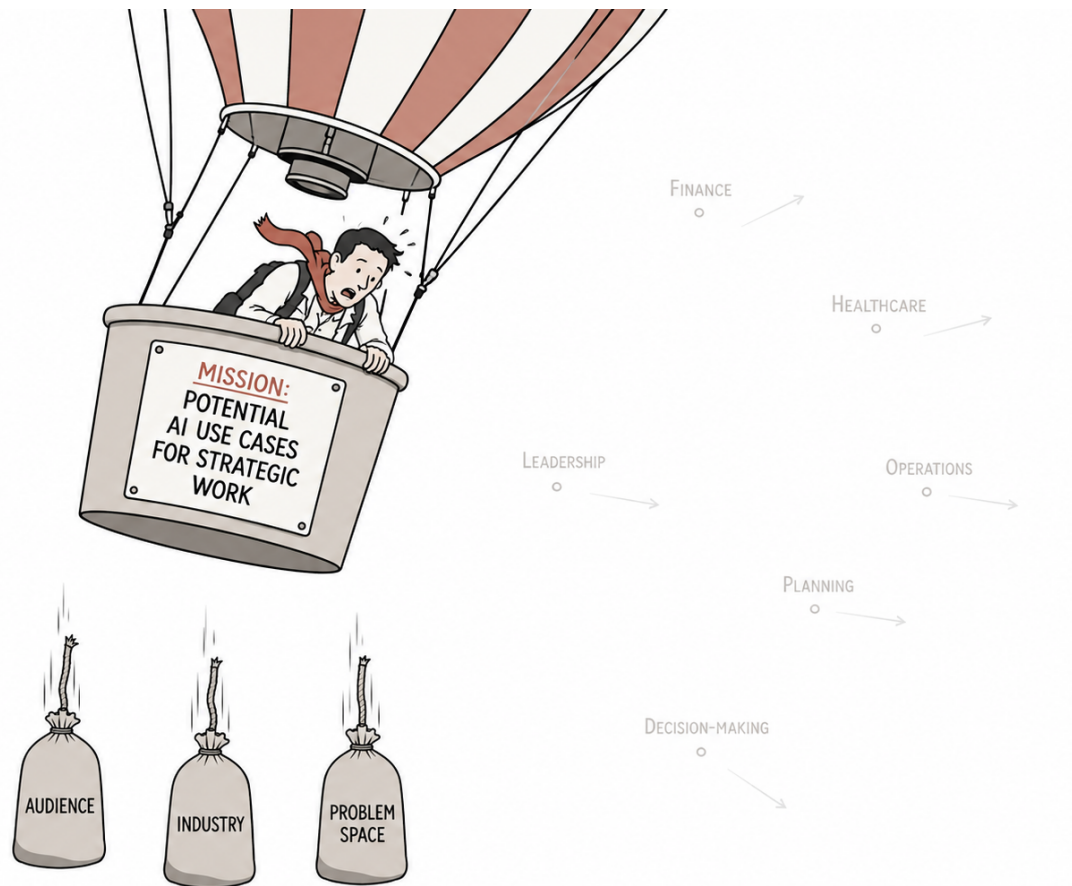


Image | Alt text: Illustration of feeling untethered | Caption: Feeling untethered without a well-defined audience, industry, or problem space. [AI-generated] | Source asset: /assets/hbs/imgConsistencyVsInterpretiveDepthTable1.png

Approach

To avoid narrowing the opportunity around our initial assumptions, I organized the work around three questions:

- How executives formulate strategy,
- What problems they encounter while doing so, and
- What role AI currently plays in their decisions.

I gathered credible resources and papers to examined these questions broadly across roles, industries, and types of strategy.

To handle research at scale, I built an AI-assisted research architecture that broke 237 papers into 992 traceable modules, translated qualitative observations into comparable measures, and preserved the evidence behind each synthesis.



Image | Alt text: Data collection | Source asset: /assets/hbs/imgImage2.png

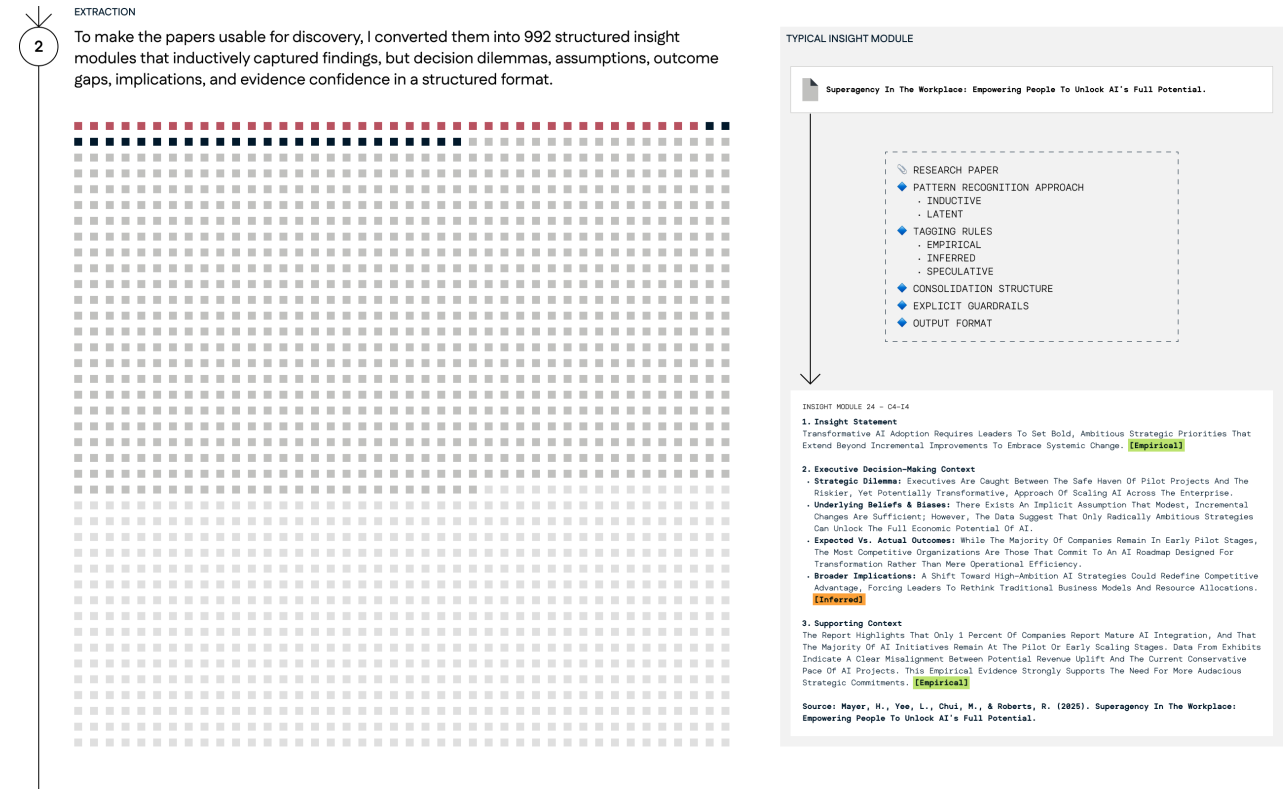
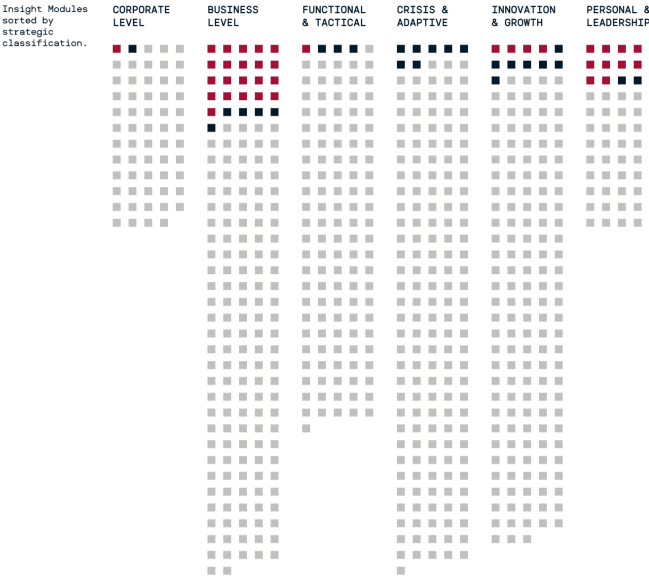


Image | Alt text: Extraction | Source asset: /assets/hbs/imgImage3.png

CLASSIFICATION

To preserve strategic context, every module was sorted by the kind of decision it described (corporate, business, functional, adaptive/crisis, innovation/growth, or personal leadership) because each setting implies different actors, stakes, time horizons, and product needs.



EXAMPLE FOR DETERMINING STRATEGY CLASSIFICATION

Insight Module 24 - C4-I4

INSIGHT MODULES

STRATEGY TYPES DESCRIPTIONS AND EXAMPLES

STRATEGY ALIGNMENT GUIDELINE

GUIDE TO USE INPUTS

TIE BREAKER RULES

OUTPUT FORMAT

EXPLICIT GUARDRAILS

STRATEGY TYPE	DECISION LAYER	STRATEGIC TENSION	STRATEGIC INTENT	SCOPE AND HORIZON	COGNITIVE FRAMING	TOTAL
CORPORATE LEVEL	2	2	2	2	2	10
BUSINESS LEVEL	3	3	3	3	3	15
FUNCTIONAL & TACTICAL	4	4	4	4	4	20
CRISIS & ADAPTIVE	1	1	1	1	1	5
INNOVATION & GROWTH	2	2	2	2	2	10
PERSONAL & LEADERSHIP	1	1	1	1	1	6

What kind of strategy does this module describe?

FUNCTIONAL & TACTICAL STRATEGY

Image | Alt text: Classification | Source asset: /assets/hbs/imgImage4.png

To prevent the synthesis from blending executive behavior with AI opportunity, I decomposed the 3 research questions into 21 binary alignment tests and used those scores to assign each module to its strongest analytical lane.

[illegible]

EXAMPLE FOR RESEARCH QUESTION GROUPING			
Insight Module 24 – C4-I4			
↓	RESEARCH QUESTION ALIGNMENT GUIDE INSIGHT MODULES GUIDE TO INTERPRETING THE INFORMATION GUIDE TO USE INPUTS TIE BREAKER RULES OUTPUT FORMAT EXPLICIT GUARDRAILS		
	RQ1	How do executives decide when to act on incomplete information versus wait for clarity?	✓
		How do executives distinguish high- from low-stakes decisions, and how does that affect process, speed, and ownership?	✗
		When strategy underperforms or conditions change, how do executives decide whether to persist, pivot, or reverse course?	✗
		What inputs most influence executive decision-making, and how do they vary by context?	✗
		How do executives balance short-term pressures with long-term priorities?	✓
		How do company size and stage shape executive decision-making and delegation?	✗
		How do industry and culture influence executive decision styles?	✗
	RQ2	What makes strategic decisions difficult, and how do executives manage uncertainty, resistance, and doubt?	✓
		What internal bottlenecks most often slow or derail strategic decisions?	✓
↓		How do executives manage trade-offs when strategic priorities conflict?	✓
		Which external pressures most shape executive decisions, and how do leaders respond when they conflict with internal priorities?	✓
		During disruption, how do executives decide whether to stay the course or pivot?	✓
		How do decision challenges differ between small companies and large enterprises?	✗
		What concerns executives most when moving from strategy to execution?	✗
	RQ3	What would make AI feel like a trusted strategic partner rather than a tool?	✗
		How have prior experiences shaped executives' views of AI for strategic decision support?	✗
		Which decisions or thought processes do executives reserve for themselves, and why?	✗
		How do executives evaluate the quality, bias, and relevance of AI-generated insights?	✗
		How should AI challenge executive assumptions without overstepping?	✗
↓		Where could AI fit naturally into executives' existing thinking workflows?	✗
		Which strategic planning frameworks are best suited to AI support, and where does AI's value break down?	✗

Image | Alt text: Inquiry routing | Source asset: /assets/hbs/img/image5.png

Maintaining Standards for AI Outputs

Reasoning models gave reliable answers but missed hidden meanings. Creativity models could go beyond basic understanding, but they weren't consistent in applying logic across different sets.

Prompt engineering brought everything together. It let us combine the creative and analytical strengths of GPT 4.5 with the steady reliability of o3. I kept the same goal but tried several versions of the AI pipeline.

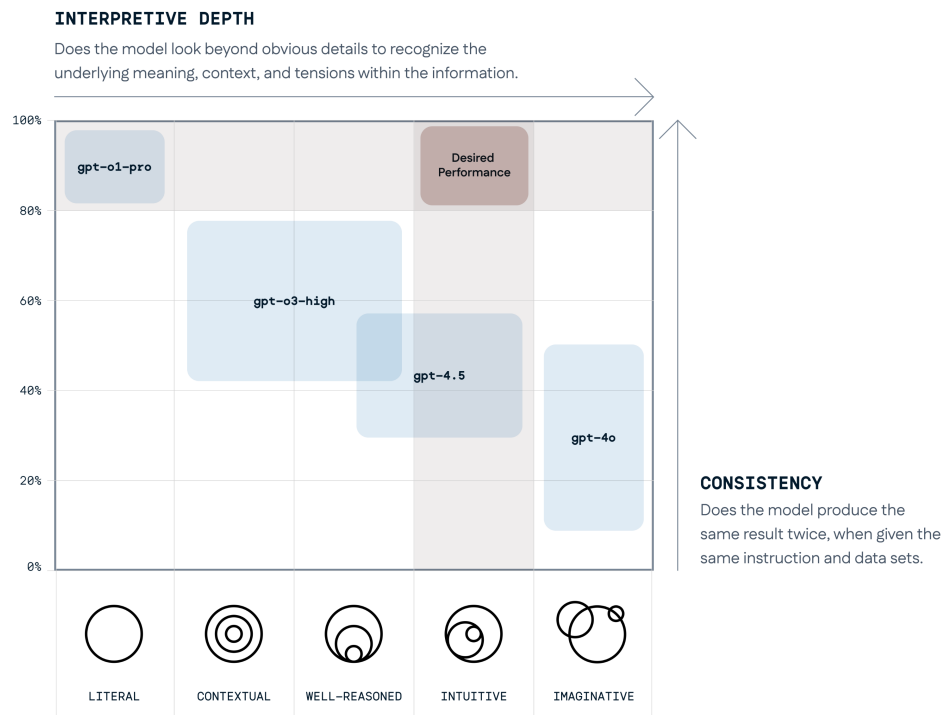


Image | Alt text: Model strengths and weaknesses mapping | Caption: Mapping current models' strengths and weaknesses against our target performance. | Source asset: /assets/hbs/imgImage69.png

Improving consistency and interpretive depth through prompting, measured using blind tests.

Iterations	gpt-4o		gpt-4.5		gpt-o3-high		gpt-o1-pro	
Iteration 01	~36%	Imaginative	~50%	Intuitive	~65%	Well-Reasoned	~95%	Literal
Iteration 02	~50% ↑	Imaginative	~62% ↑	Intuitive	~65%	Well-Reasoned	~95%	Literal
Iteration 03	~40% ↓	Imaginative	~54% ↓	Imaginative ↑	~75% ↑	Well-Reasoned	🛑 stopped exploration	
Iteration 04	~45% ↑	Imaginative	~72% ↑	Well-Reasoned ↓	~85% ↑	Well-Reasoned		
Iteration 05	🛑 stopped exploration		~81% ↑	👉 Intuitive ↑	~85%	Well-Reasoned		
Iteration 06			~76% ↓	Well-Reasoned ↓	🛑 stopped exploration			
Iteration 07			~88% ↑	Well-Reasoned				
Iteration 08			~72% ↓	Well-Reasoned				

Consistency Improved.

But, the interpretation remained too abstract and imaginative.

Required multiple iterations to identify the perfect balance between consistency and interpretive depth.

While consistency could be improved, interpretations remained 'Well-Reasoned' but not 'Intuitive'.

gpt-o1 helped establish a benchmark for consistency, but improving interpretive depth proved challenging.

Image | Alt text: Consistency and interpretive depth model comparison | Caption: Models were compared by consistency and interpretive depth to identify the strongest balance between reliable output and meaningful interpretation. | Source asset: /assets/hbs/imgImage71.png

Designing Situational Axes.

Despite multiple attempts at sorting, the resulting clusters were still too large to facilitate meaningful collaboration or productive discussion.

I considered subdividing the clusters by industry; however, this approach risked reinforcing industry stereotypes. Activities within a single industry can differ significantly. For example, marketing within pharmaceuticals and legal operations within a content generation team represent distinct functions.

To address this challenge, I developed a novel approach for mapping activities. Each activity was evaluated along shared dimensions, including regulation, modularity, timing, value horizon, knowledge transfer, and market spread, using situational axes.

This approach ensured that each collection was mutually exclusive and collectively exhaustive, allowing every module to be mapped without exception.

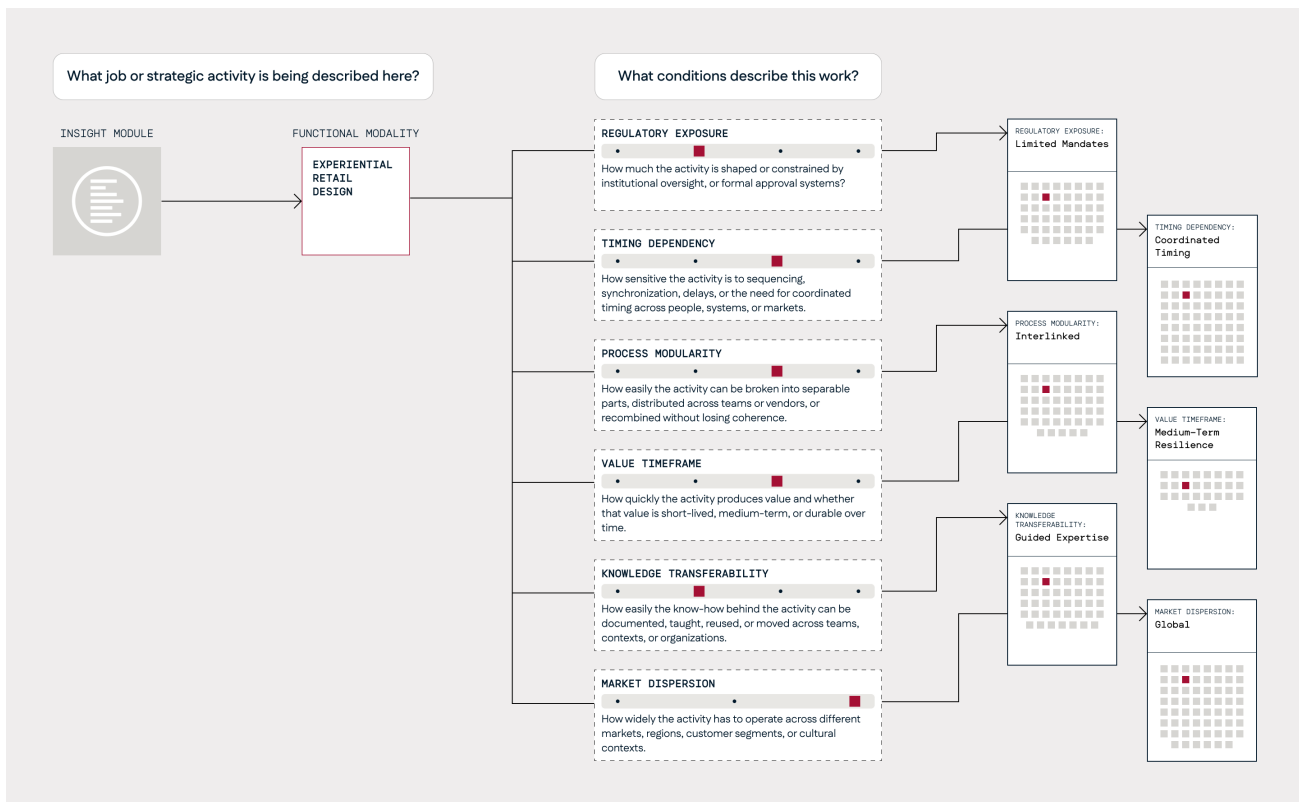


Image | Alt text: Situational tags applied to research modules | Caption: Tagging each module based on the situations influencing the activity. | Source asset: /assets/hbs/imgPanwImage012.png

With clusters finally manageable, I used contrastive typology: scoring evidence along bipolar situational dimensions and comparing their polar intersections to identify distinct operating contexts.

Although the dimensions produced several plausible permutations, collaborative working sessions helped us eliminate factors that did not generate meaningful distinctions, felt insufficiently relevant to the team's objectives, or were unlikely to support useful product decisions.



Image | Alt text: Four situational axes | Caption: Four situational axes created a common language for comparing otherwise unrelated forms of strategic work. | Source asset: /assets/hbs/imgImage32.png

Approach 1: Selecting a theme

Here you can chose to address different archetypes, for a particular use case.

Recommendation: Technology and Human Tension.

People 🧑	Legacy Navigators: Designs enduring systems within slow-moving, legacy-laden environments, prioritizing future relevance over speed.	Urgent Executioners: Acts urgently within rigid boundaries, balancing high-speed delivery with tightly enforced external constraints.	Coordinated Builders: Thrives in modular yet interdependent systems, where value depends on precisely timed, cross-functional coordination.	Principled Orchestrators: Navigates long-horizon complexity through layered decision-making, emphasizing principled clarity over rapid action.
Themes 🧩				
Relational Fragility: The struggle to grow while building trust and collaboration, potentially weakening execution and capability.	Scale 🧑 Customization	Short-Term Gains 🧑 Long-Term Capability	Operational Independence 🧑 Strategic Partnerships	Psychological Safety 🧑 Hierarchical Dynamics
Underestimated Complexity: Leaders misjudging complexity and struggling to build a strategy for changing landscapes.	Transformation 🧑 Legacy Systems	Speed 🧑 Constraints	Internal Identity 🧑 Aggressive Growth	Intuitive Decisions 🧑 Structured Analytics
Emotional Barriers to Change: Challenges while bringing change, recalibration, and experimentation.	Emotional Resonance 🧑 Digital Innovation	Strategic Narratives 🧑 Evolving Realities	Proven Models 🧑 Disruptive Pricing	Cognitive Flexibility 🧑 Cognitive Inertia
Technology and Human Tension: Evolution of technology outpacing governance, trust, and human judgement safeguards.	Predictive Analytics 🧑 Human Intuition	Deployment 🧑 Governance	Craftsmanship 🧑 Brand Perception	Ethical Integrity 🧑 AI-Driven Efficiency

Image | Alt text: View slide 1 | Source asset: /assets/hbs/imgImage79.png

Approach 2: Selecting an archetype

Here you can chose to address different use cases, for a single archetype.

Recommendation: Coordinated Builders

People 🧑	Legacy Navigators: Designs enduring systems within slow-moving, legacy-laden environments, prioritizing future relevance over speed.	Urgent Executioners: Acts urgently within rigid boundaries, balancing high-speed delivery with tightly enforced external constraints.	Coordinated Builders: Thrives in modular yet interdependent systems, where value depends on precisely timed, cross-functional coordination.	Principled Orchestrators: Navigates long-horizon complexity through layered decision-making, emphasizing principled clarity over rapid action.
Themes 🧩				
Relational Fragility: The struggle to grow while building trust and collaboration, potentially weakening execution and capability.	Scale 🧑 Customization	Short-Term Gains 🧑 Long-Term Capability	Operational Independence 🧑 Strategic Partnerships	Psychological Safety 🧑 Hierarchical Dynamics
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Approach 3: a T shaped approach

Recommendation: Coordinated Builders + Technology and Human Tension.

Caution: Each problem statement is a massive design scope.
A T-shaped approach could potentially explode your scope.

People🧑 Themes🧠	Legacy Navigators: Designs enduring systems within slow-moving, legacy-laden environments, prioritizing future relevance over speed.	Urgent Executioners: Acts urgently within rigid boundaries, balancing high-speed delivery with tightly enforced external constraints.	Coordinated Builders: Thrives in modular yet interdependent systems, where value depends on precisely timed, cross-functional coordination.	Principled Orchestrators: Navigates long-horizon complexity through layered decision-making, emphasizing principled clarity over rapid action.
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Technology and Human Tension: Evolution of technology outpacing governance, trust, and human judgement safeguards.	Predictive Analytics🧑 Human Intuition	Deployment🧑 Governance	Craftsmanship🧑 Brand Perception	Ethical Integrity🧑 AI-Driven Efficiency

Image | Alt text: View slide 3 | Source asset: /assets/hbs/imgImage81.png

Approach 4: Selecting an individual problem

Here, you narrow down on one out of many problem statements that resonates most with you.

Recommendation:
Craftsmanship vs. Brand Perception (or) Proven Models vs. Disruptive Pricing.

People🧑 Themes🧠	Legacy Navigators: Designs enduring systems within slow-moving, legacy-laden environments, prioritizing future relevance over speed.	Urgent Executioners: Acts urgently within rigid boundaries, balancing high-speed delivery with tightly enforced external constraints.	Coordinated Builders: Thrives in modular yet interdependent systems, where value depends on precisely timed, cross-functional coordination.	Principled Orchestrators: Navigates long-horizon complexity through layered decision-making, emphasizing principled clarity over rapid action.
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Image | Alt text: View slide 4 | Source asset: /assets/hbs/imgImage82.png

I used the matrix to show HBS four ways to narrow the opportunity for further discovery—by recurring problem, operating context, a combination of both, or one focused starting point.

Designing Principles of Human-AI interaction

While not part of the scope, I recognized we had rich data, capable of helping us define design principles for AI interactions among executives for strategic discussions.

This allowed us to test different scenarios and contexts, helping us fine-tune the voice of an AI system designed specifically for executives.

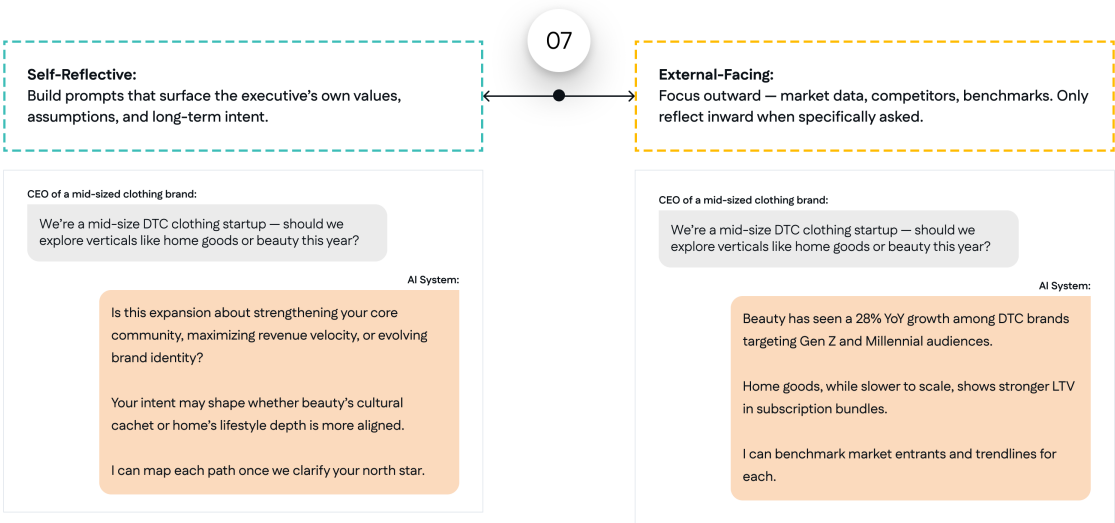


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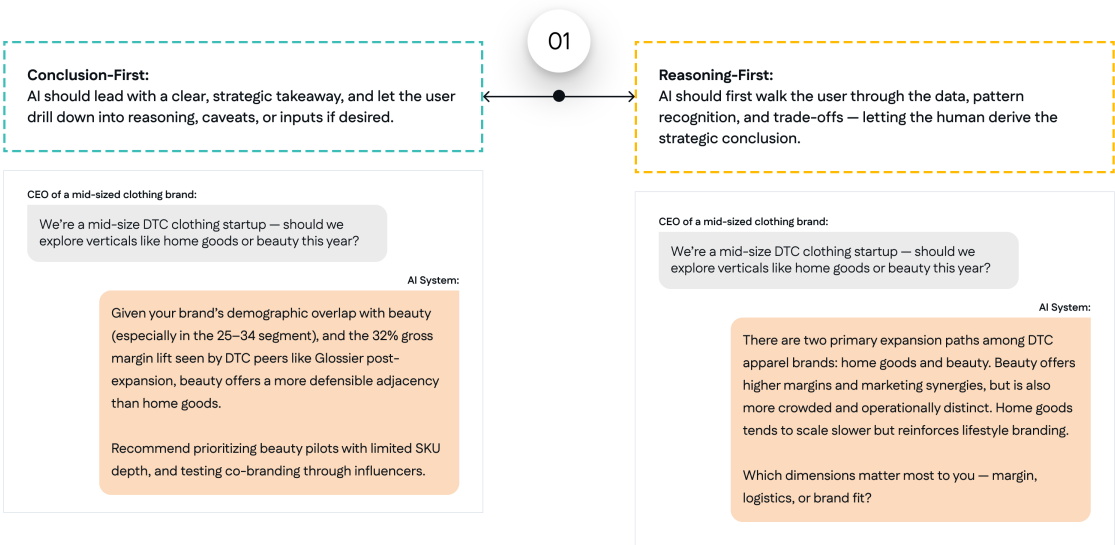


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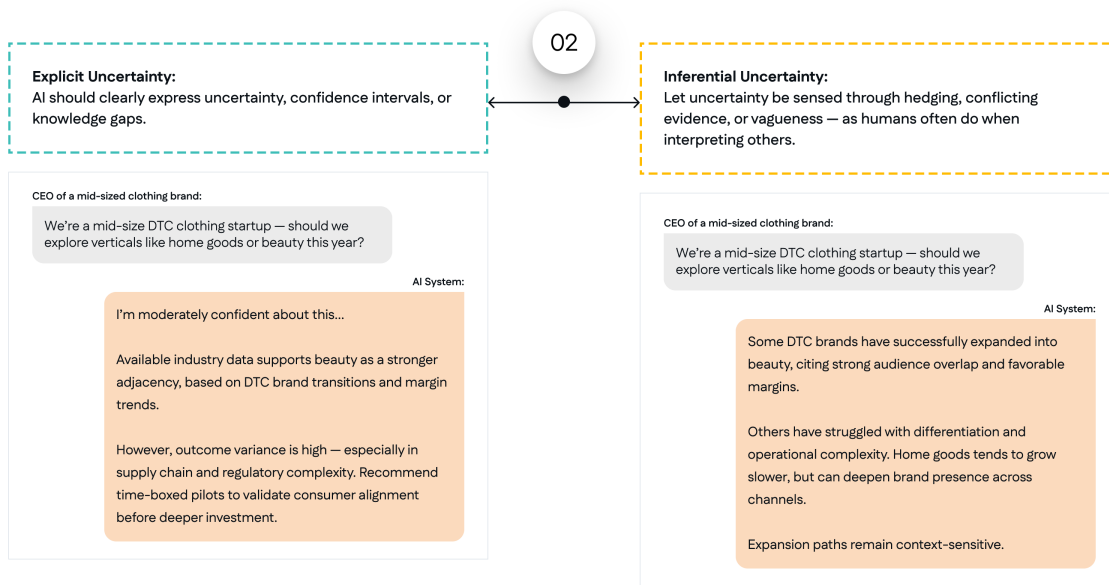


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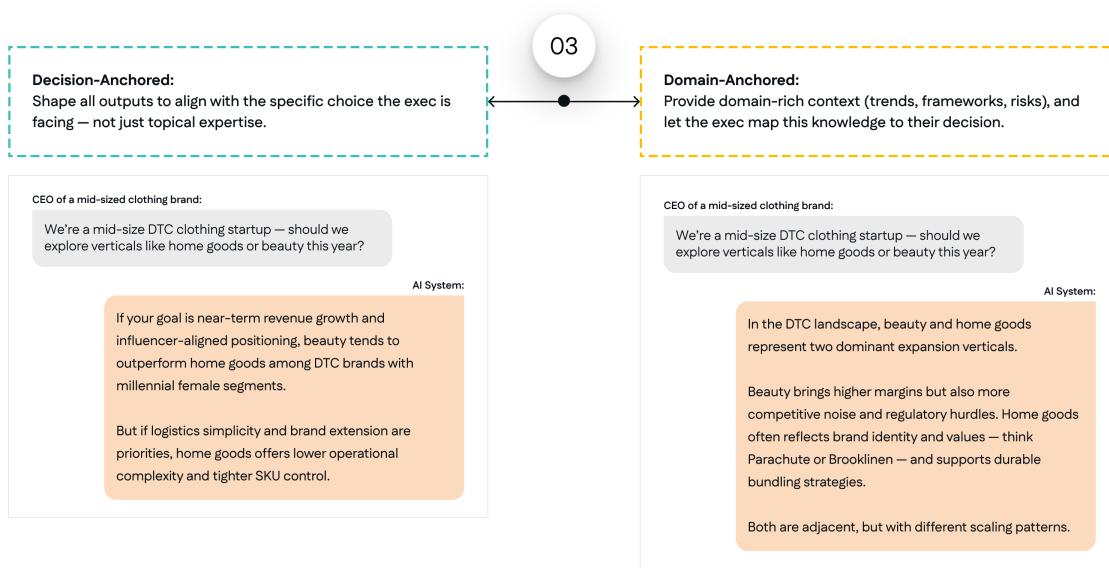


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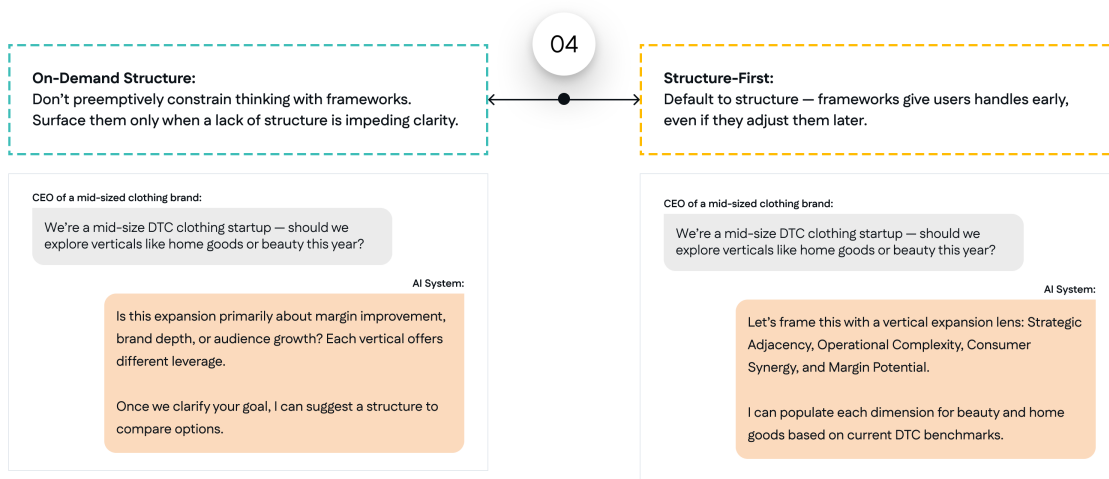


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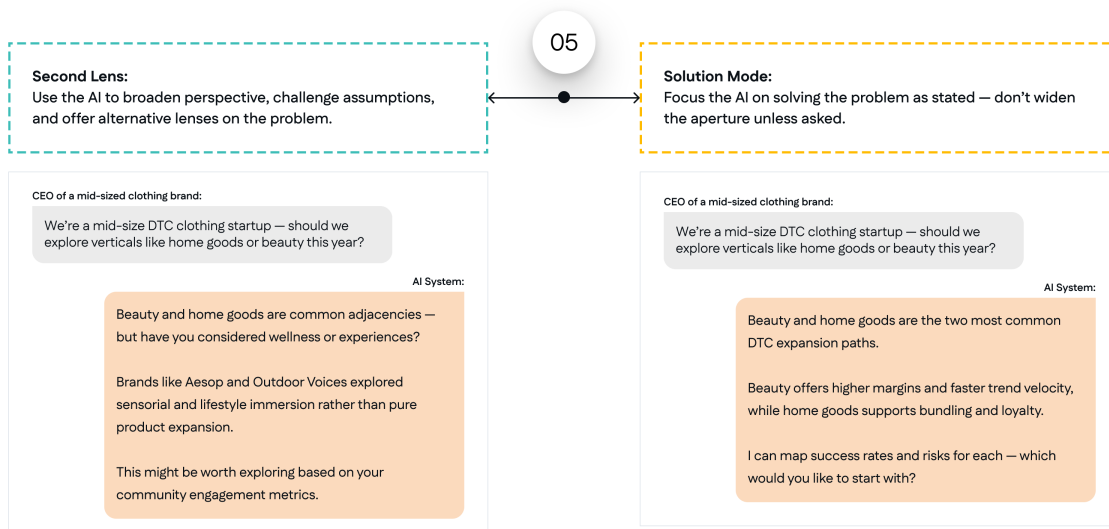


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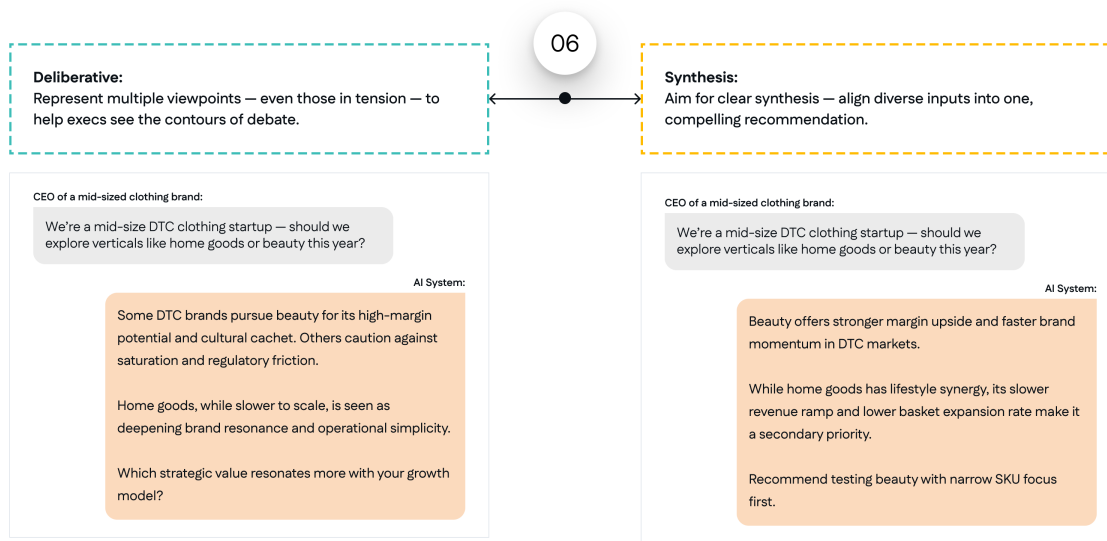


Image | Alt text: View slide 7 | Source asset: /assets/hbs/imgImage77.png

Design principles often sound good but aren't actually useful. When they become truisms like "AI should be trustworthy," "prioritize the user," or "support human judgment," they're easy to agree with but hard to use in real design work.

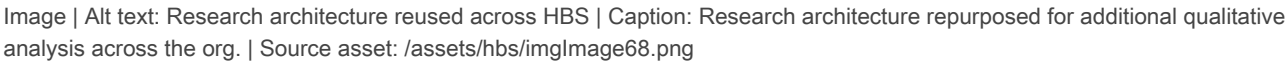
Having many modules in Research Question 3 let us dig deeper and find hidden meanings. This helped me develop a way to frame principles as thoughtful trade-offs, grounded in context rather than as absolute truths.

I wrote each principle so its opposite could also make sense; something a smart team might choose in another situation or with different goals.

Impact:

The work continued in three ways.

- The seven human-AI interaction principles were used to fine-tune model responses across exploratory HBS projects for business executives. They gave teams specific behaviors to compare. For example, whether a model should present its conclusion or reasoning first, accept the user's framing or introduce another perspective, and state uncertainty directly or leave it implicit.
- The research architecture was repurposed for other qualitative work. We used it whenever it was necessary to convert unstructured text into comparable measures and reviewable patterns. Two immediate examples included scoring participant-written prompts at scale for HBS' Sana.ai prompt engineering course, and analyze open-ended surveys from Leading with AI events.
- The sixteen people problems initiated follow-on research and exploration across HBS teams. Instead of beginning with "executive strategy" as one enormous subject, teams could investigate specific tensions such as deployment versus governance, short-term gains versus long-term capability, or intuitive judgment versus structured analysis.



Once hundreds of sources moved through the study, prompts, definitions, scoring rules, model choices, and review checkpoints became part of the research method.